

Multicast SFC Embedding in Software-defined SAGIN with Heterogeneous Network Resources

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Abstract—Space-Air-Ground Integrated Network (SAGIN) is emerged as a promising paradigm to realize the vision of global coverage of sixth generation (6G) communication network. As an efficient communication pattern, multicast communication can be widespread among the ever increasing communication requests in the seamless access SAGIN. However, given the extensive network scale, time varying characteristic and heterogeneity of SAGIN, the coordination of heterogeneous networks brings challenges to resource allocation. With virtualization technologies, the physical resources within SAGIN can be virtualized as Virtual Network Functions (VNFs) in virtual resource pool. To ensure that multicast services can be provided, the network deploys Multicast Service Function Chains (MSFCs) where the service flows is processed by VNFs in order during the transmission. In this paper, we study the problem of jointly optimizing VNF placement and multicast routing within SAGIN while considering that different network segments are equipped with different physical network resources and cost coefficients. We formulate it as an Integer Linear Programming (ILP) problem with the objective of maximizing network revenue while minimizing resource cost of computation and bandwidth. Since it is NP-hard, a low complexity heuristic algorithm is developed to find an efficient solution within polynomial time. Simulation results demonstrate the effectiveness of our proposed model and algorithm, and the deployment cost and blocking rate can be significantly reduced in SAGIN compared with independent terrestrial network.

Index Terms—Space-air-ground integrated network (SAGIN), multicast communication, virtual network function (VNF), multicast service function chain (MSFC).

I. INTRODUCTION

In the future era of sixth generation (6G) communication network, the emerging applications, such as holographic communication, the industrial Internet of Things (IoT), ubiquitous Artificial Intelligence (AI) [1], etc., will necessitate a wide array of multicast services including high definition streaming, system supervision and upgrade, edge learning [2] and so on. However, the current terrestrial mobile networks are unable to meet the diverse service requirements for seamless access, ubiquitous coverage, and low-latency communication. In recent years, the promising network architecture called Space-Air-Ground Integrated Network (SAGIN) is proposed

to satisfy such requirements. It integrates satellite, Unmanned Aerial Vehicle (UAV), High Altitude Platform (HAP) and terrestrial communication networks, leveraging the wide coverage capability of satellites and airborne platforms and the well-developed infrastructural facilities of terrestrial communication networks [3].

However, SAGIN encompasses various types of communication devices and dynamic network topology, creating complexities for network operation managements [1]. Recently, software-defined SAGIN architecture is proposed [4]–[6], which leverages Network Function Virtualization (NFV) and Software-Defined Networking (SDN). NFV virtualizes traditional network equipment into Virtual Network Functions (VNFs) [7], enabling the coordination of heterogeneous network resources within SAGIN. Ensuring secure, fast, and stable multicast services, SDN controller implements Multicast Service Function Chains (MSFCs) orchestration, where the service flows are processed by VNFs in a predetermined order during intermediate transmission [8]. Compared to unicast communication, multicast saves network bandwidth and computing resources as it avoids sending redundant data packets during transmission.

It is challenging to embed MSFCs within SAGIN. First, The placement of VNF instances influences traffic routing, determining both computation and communication resource consumption. Reusing VNFs may expand multicast function trees, leading to prohibitive communication costs. Conversely, efficient routing with minimal hops between source and each destination may install more VNF instances to meet Service Function Chain (SFC) constraints. Second, unlike terrestrial networks where nodes are typically homogeneous, the network resources on different nodes within SAGIN vary in value, with aerial crafts having more limited and precious resources than ground nodes. Relying on space or airborne nodes to reduce long-distance communication hops for meeting latency requirements may incur high resource costs.

Recent studies on MSFCs embedding are focused on the terrestrial network scenarios [9]–[11], while the works in non-terrestrial network scenarios are concentrated on unicast communication [5], [6]. Nevertheless, there is a lack of re-

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logical link set, where h_i represents a logical link connecting VNF v_{i-1} to v_i , and it can be mapped onto multiple physical links. ϵ signifies the bandwidth requirement of r . $L = \{l_1, l_2, \dots, l_{|V|+1}\}$ denotes maximum tolerable latency set for the logical links. $\Phi = \{\phi_1, \phi_2, \dots, \phi_{|D|}\}$ represents network revenue set generated by each destination user, where ϕ_i is calculated according to the average minimum hop count matrix of network topology.

III. PROBLEM FORMULATION

A. Binary Decision Variables

We define five binary decision variables to model the MSFC embedding problem as an ILP problem. The variable δ_d takes a value as 1 if the request required by user d is served and 0 otherwise. The variable $x_{v,n,d}$ indicates whether the VNF v required by the user at destination d is deployed on physical node n or not. The variable $y_{h,e,d}$ donates whether the logical link h of the user at destination d is mapped onto the physical link e or not. The variable $z_{v,n}$ is used to determine whether the VNF v required by the multicast request is embedded on physical node n or not. The variable $w_{h,e}$ indicates whether the logical link h of the multicast request is mapped onto the physical link e or not. It should be noted that $x_{v,n,d}$ and $y_{h,e,d}$ are considered from the perspective of each destination user, whereas $z_{v,n}$ and $w_{h,e}$ are defined for the whole multicast flow. Therefore, they have the following relationships:

$$x_{v,n,d} \leq z_{v,n}, \quad \forall d \in D, v \in \hat{V}, n \in N, \quad (2)$$

$$y_{h,e,d} \leq w_{h,e}, \quad \forall d \in D, h \in H, e \in E. \quad (3)$$

B. Deployment Costs

To comprehensively reflect the varying scarcity of network resources in heterogeneous SAGIN, we define physical network resource cost model for different network segments. The total deployment cost C_{total} is given by:

$$C_{total} = \sum_{e_{n,m} \in E} \sum_{h \in H} \varphi_{e_{n,m}}^{cm} \cdot \epsilon \cdot w_{h,e} + \sum_{v_i \in \hat{V}} f_{v_i} \cdot (\varphi_G^{cp} \sum_{n \in N_G} z_{v_i,n} + \varphi_A^{cp} \sum_{n \in N_A} z_{v_i,n} + \varphi_S^{cp} \sum_{n \in N_S} z_{v_i,n}), \quad (4)$$

where $\varphi_{e_{n,m}}^{cm}$ is the cost per unit bandwidth required for link $e_{n,m}$. For the sake of simplification, $\varphi^{cm} = \{\varphi_{e_{n,m}}^{cm} | \varphi_{e_{n,m}}^{cm} \in \varphi^{cm}\}$ includes the unit cost for each link set. $\varphi^{cp} = \varphi_G^{cp} \cup \varphi_A^{cp} \cup \varphi_S^{cp}$, where φ_G^{cp} , φ_A^{cp} , and φ_S^{cp} represent the cost per unit computational resource consumption for ground, aerial, and satellite nodes, respectively.

C. Objective Function

Our objective is to maximize the sum of network revenues generated by destination users while minimizing the total network resource cost. It should be noted that providing services for more users implies more users paying to the service provider. However, when deployment costs outweigh revenues, the SDN controller will reject the deployment strategy even if it is technically feasible without considering costs. The

problem of jointly optimizing VNF placement and multicast routing within SAGIN is formulated as follows:

$$\text{Maximize} \quad \sum_{d \in D} \phi_d \cdot \delta_d - C_{total}, \quad (5)$$

$$\text{s.t.} \quad (2) - (3),$$

$$x_{v_0,s,d} = \delta_d, \quad \forall d \in D, \quad (6)$$

$$x_{v_{|V|+1},d,d} = \delta_d, \quad \forall d \in D, \quad (7)$$

$$\sum_{n \in N} x_{v,n,d} = \delta_d, \quad \forall d \in D, v \in \hat{V}, \quad (8)$$

$$\sum_{m \in N} y_{h,e_{n,m},d} \leq 1, \quad \forall n \in N, d \in D, h \in H, \quad (9)$$

$$\sum_{m \in N} y_{h,e_{m,n},d} \leq 1, \quad \forall n \in N, d \in D, h \in H, \quad (10)$$

$$\sum_{m \in N} y_{h_j,e_{n,m},d} - \sum_{m \in N} y_{h_j,e_{m,n},d} = x_{v_{j-1},n,d} - x_{v_j,n,d}, \quad \forall d \in D, h_j \in H, v_j \in \hat{V}, n \in N, \quad (11)$$

$$\sum_{v_i \in \hat{V}} f_{v_i} \cdot z_{v_i,n} \leq C_N(n), \quad \forall n \in N, \quad (12)$$

$$\sum_{h \in H} \epsilon \cdot w_{h,e} \leq B(e), \quad \forall e \in E, \quad (13)$$

$$\sum_{e \in E} y_{h,e,d} \cdot \tau_e \leq l_h, \quad \forall d \in D, h \in H. \quad (14)$$

In the objective function (5), the first term represents sum of network revenues. Constraints (6) and (7) indicate that the service flow admitted to each destination must start from source node and terminate at the destination node. Constraint (8) implies that the service flow admitted to each destination requires that its VNF v can be deployed on one and only one physical node, ensuring that the service flow is processed by VNF v only once. Constraints (9) and (10) state that the service flow admitted to each destination is not allowed to be split into multiple flows. Constraint (11) indicates flow conservation. Constraints (12) and (13) ensure that the computation and bandwidth resource consumption can not exceed the capacity of the nodes and links, respectively. Constraint (14) restricts the delay of each logical link flow from exceeding the maximum tolerable latency.

IV. HEURISTIC ALGORITHM

The joint VNF placement and multicast routing problem can be proved that it is NP-hard, as its special case, that the traditional multicast problem without SFC requirements is NP-hard [12]. As the problem scale increases, existing solvers can hardly find optimal solutions within polynomial time. In order to efficiently solve the problem in a shorter time, we develop a heuristic algorithm, the Resource Scarcity-Awared MSFC Embedding Algorithm (RSA).

As shown in Algorithm 1, RSA starts with clustering destination nodes. It identifies the central destination node among the destination nodes based on the delay matrix O_{delay} , ensuring relatively uniform latency distributions from the central

Algorithm 1 Resource Scarcity-Awared MSFC Embedding Algorithm(RSA)

Input: $G(N, E), r = \{s, D, V, H, \epsilon, L, \Phi\}$.

Output: MSFC embedding strategy of the service flow.

- 1 Initialize B, C_N, O_{delay} according to network parameters;
 - 2 $d_c \leftarrow Cluster(D, O_{delay})$;
 - 3 $D' \leftarrow SortAscending(O_{delay}(d_c, D))$;
 - 4 **for** $d \in D'$ **do**
 - 5 Search embedding strategy of $s - d$ pair by Algorithm 2;
 - 6 Calculate the network net revenue according to the current strategy;
 - 7 Update B, C_N according to the current strategy;
-

node to each destination node. Subsequently, the destination nodes are sorted in ascending order according to the latency. Furthermore, for each destination node, Stage Strategy Procedure (SSP) is used to find the deployment strategy from source nodes to each destination node, as shown in Algorithm 2. Finally, SSP calculate the network net revenue according to the current strategy.

In the subsequent sections, we will provide a comprehensive explanation of SSP. SSP comprises $|V| + 2|$ stages, which encompass the deployment of real VNFs and the placement of virtual VNFs at both the source and destination nodes. In each stage, SSP initially determines the current VNF placement location. Then, it proceeds to search for a routing path between the previous VNF placement location and the current one, within expanded service node set $\hat{N}_{ser}^i$, as shown in (15).

$$\hat{N}_{ser}^i = \begin{cases} s & i = 0, \\ N_{ser} & 1 \leq i \leq |V|, \\ d & i = |V| + 1. \end{cases} \quad (15)$$

In the i -th stage, first, SSP deploys VNF v_i on node $a \in \hat{N}_{ser}^i$, recording the weighted computational resource consumption $\theta_{a',a}^{i,comp}$, which is calculated by (16).

$$\theta_{a',a}^{i,comp} = \begin{cases} f_{v_i} \cdot \varphi^{cp}(a) & 0 < f_{v_i} < C_{N,a'}^{i-1}(a), \\ \infty & \text{otherwise.} \end{cases} \quad (16)$$

Second, it traverses $\hat{N}_{ser}^{i-1}$ which is the location of v_{i-1} to find routing paths between a and a' by K-shortest path Algorithm according to $Q_{Lcost}^{i,a'} = \{q^{i,a'}(e_{n,m}) | a' \in N^{i-1}, n, m \in N\}$. $Q_{Lcost}^{i,a'}$ is the weighted bandwidth cost matrix and computed according to (17).

$$q^{i,a'}(e_{n,m}) = \begin{cases} \epsilon \cdot \varphi_{e_{n,m}}^{cm} & \epsilon < B_{a'}^{i-1}(e_{n,m}), \\ \infty & \text{otherwise.} \end{cases} \quad (17)$$

Third, SSP check the delay of each path obtained by K-shortest path Algorithm. If there exists one path satisfies the delay constraint among these K paths, the path and its weighted bandwidth consumption are recorded as $p_{a',a}^i$ and $\theta_{a',a}^{i,comm}$. In cases where no path meets the delay constraint, the Dijkstra Algorithm is used to find the shortest delay path according to the shortest delay matrix O_{delay} . If this path satisfies the logical link delay constraint, return $p_{a',a}^i$ and $\theta_{a',a}^{i,comm}$. Finally, the total cost $\theta_{a',a}^i$ in stage i from node a' to node a is calculated by $\theta_{a',a}^i = \theta_{a',a}^{i,comp} + \theta_{a',a}^{i,comm}$.

Algorithm 2 Stage strategy Procedure(SSP)

- 1 Initialize VNFs location $\Omega = \emptyset$, Multicast tree $\Upsilon = \emptyset$;
 - 2 Initialize Q_{Lcost} from (17);
 - 3 **for** $i = 1 : |V| + 1$ **do**
 - 4 **for** $a \in \hat{N}_{ser}^i$ **do**
 - 5 **for** $a' \in \hat{N}_{ser}^{i-1}$ **do**
 - 6 Calculate $\theta_{a',a}^{i,comp}$ from (16) ;
 - 7 $(P, C_{comm}) \leftarrow Kshortestpath(Q_{Lcost}^{i,a'}, a', a)$;
 - 8 **for** $p \in P$ **do**
 - 9 $p_{delay} \leftarrow$ Calculate total delay of path p ;
 - 10 **if** $p_{delay} \leq l_i$ **then**
 - 11 $p_{a',a}^i \leftarrow p, \theta_{a',a}^{i,comm} \leftarrow C_{comm}(p_{a',a}^i)$;
 - 12 **else**
 - 13 **break** ;
 - 14 **if no path satisfy the time constraint then**
 - 15 $(delay, p_{a',a}^i) \leftarrow Dijkstra(O_{delay}, a', a)$;
 - 16 Calculate $\theta_{a',a}^{i,comm}$ of $p_{a',a}^i$ if $delay \leq l_i$;
 - 17 $\theta_{a',a}^i = \theta_{a',a}^{i,comp} + \theta_{a',a}^{i,comm}$;
 - 18 $\lambda_a^i = \min_{a' \in \hat{N}_{ser}^{i-1}} \{\lambda_{a'}^{i-1} + \theta_{a',a}^i\}$;
 - 19 $\hat{a} = \arg \min_{a' \in \hat{N}_{ser}^{i-1}} \{\lambda_{a'}^{i-1} + \theta_{a',a}^i\}$;
 - 20 $\Omega_a^i \leftarrow (\Omega_{\hat{a}}^{i-1}, a), \Upsilon_a^i \leftarrow (\Upsilon_{\hat{a}}^{i-1}, p_{\hat{a},a}^i)$;
 - 21 Get B^i, C_N^i by updating $B_{\hat{a}}^{i-1}, C_{N,\hat{a}}^{i-1}$;
 - 22 Get $Q_{Lcost}^{i+1,a}$ from (17) when $i < |V| + 1$;
 - 23 **if** $\lambda_a^i = \infty, \forall a \in N^i$ **then**
 - 24 **return** $\Omega \leftarrow \emptyset, \Upsilon \leftarrow \emptyset, B, C_N$;
 - 25 $a^* \leftarrow \arg \min_{a \in N^{|V|+1}} \lambda_a^{|V|+1}$;
 - 26 Update B, C_N according to $B_{a^*}^{|V|+1}, C_{N,a^*}^{|V|+1}$;
 - 27 **return** $\Omega \leftarrow \Omega_{a^*}^{|V|+1}, \Upsilon \leftarrow \Upsilon_{a^*}^{|V|+1}, B, C_N$;
-

After all nodes in $\hat{N}_{ser}^{i-1}$ are traversed, the accumulative cost λ_a^i is calculated according to $\lambda_a^i = \min_{a' \in \hat{N}_{ser}^{i-1}} \{\lambda_{a'}^{i-1} + \theta_{a',a}^i\}$. Then, $\hat{a} = \arg \min_{a' \in \hat{N}_{ser}^{i-1}} \{\lambda_{a'}^{i-1} + \theta_{a',a}^i\}$ is used to find the node $\hat{a}$ with the minimum weighted consumption cost. Ω_a^i is constructed from the previous stage deployment strategy $\Omega_{\hat{a}}^{i-1}$ and deploying v_i on node a . Similarly, Υ_a^i is constructed from the previous stage routing strategy $\Upsilon_{\hat{a}}^{i-1}$ and $p_{\hat{a},a}^i$. After that, SSP updates B^i and C_N^i according to $B_{\hat{a}}^{i-1}$ and $C_{N,\hat{a}}^{i-1}$. Besides, SSP get $Q_{Lcost}^{i+1,a'}$ from (17) if it is not the last stage.

The above steps are repeated for each stage. Once all the accumulative cost λ_a^i is infinity, the branch to destination node d of multicast tree is rejected, and SSP terminates. If all stages are successfully deployed, SSP selects the service node a^* with the minimum $\lambda_{a^*}^{|V|+1}$ and updates B and C_N . Finally, SSP returns the placement strategy and routing path and B and C_N .

Now we analyze the complexity of RSA. Cluster Algorithm finds the center destination and sort D can be achieved in $O(|D|^2 + |D| \log(|D|))$. In SSP, constructing Q_{Lcost} needs $O(|N|^2)$ steps. K-shortest path Algorithm requires $O(2|E| + |N| \log |N| + K)$ and checking the path delay to determine whether to use Dijkstra Algorithm is in $O(K|E| + |N|^2 + |E|)$. Besides, updating B^i and C^i requires $O(|N| + |E|)$ steps. Therefore, SSP needs $O(|N_{ser}|^2(|N|^2 + K|E|))$. Since $|D| <$

TABLE I
NETWORK PARAMETERS

Parameter	Value	Parameter	Value
N	{50, 100}	φ_{G}^{cp}	10
$N_G : N_A : N_S$	6: 3: 1	φ_{A}^{cp}	20
$C_N(N_G)$	U[50, 100]	φ_{S}^{cp}	30
$C_N(N_A)$	U[50, 80]	φ_{EG}^{cm}	5
$C_N(N_S)$	U[50, 80]	φ_{EA}^{cm}	20
$B(E_S)$	U[50, 100]	φ_{ES}^{cm}	30
$B(E_A)$	U[50, 80]	φ_{EG-A}^{cm}	20
$B(E_S)$	U[50, 80]	φ_{EG-S}^{cm}	30
$B(E_{G-A})$	U[50, 80]	φ_{EA-S}^{cm}	30
$B(E_{G-S})$	U[50, 80]	N_{ser}	20
$B(E_{A-S})$	U[50, 80]	$ V $	5
VNF	U[1, 50]	ϵ_r	U[1, 50]
$\tau_{em,n}$	P(10)	R	1

$|N|$, the time complexity of RSA is $O(|D||V_r||N_{Ser}|^2(|N|^2 + K|E|))$.

V. PERFORMANCE EVALUATION

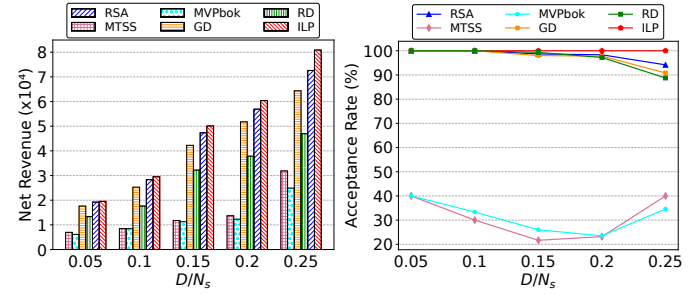
A. Parameter Setting and Simulation Environment

We consider the quasi-static network topology in a certain time period, and construct a physical network to simulate SAGIN by programming. The network parameters are as shown in Table I [5], [13]. All simulations are repeated 30 times and reported the average in a server with Intel(R Core(TM)2 Duo CPU T7700 (2.4GHz)) processors and 128GB of RAM. The optimal solution named ILP, which is obtained by ILP solver.

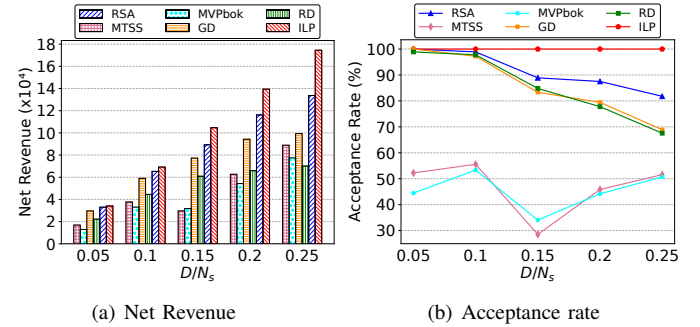
B. Comparison Algorithms

As previously mentioned, there has been no exist researches on deploying MSFC in SAGIN. Therefore, we have designed two variant algorithms based on our approach, Greedy Deployment (GD) strategy and Random Deployment (RD) strategy. GD does not require traversing all service nodes, it decides the deployment of VNF v_i based only on the cumulative cost built in the i -th stage instead. RD randomly selects service nodes for cumulative cost construction, rather than traversing from ground nodes as in SSP and GD, which is an extension of the GD algorithm.

Additionally, we adopt the Multicast VNF Placement based on K-Shortest Path (MVPboK) and Multicast Tree with Source Selection (MTSS) Algorithms in [9] for comparison, whose steps for finding candidate sources are omitted, as we focus on single source scenario. Besides, both of them ignore the influence of heterogeneous network resources. For MVPboK, it initially selects the central destination node and then finds K shortest paths from the source node to this central node. Subsequently, it searches for a path that accommodates the deployment of all VNFs among the paths. Finally, it constructs a multicast tree by considering the shortest path and VNFs placement, aiming to maximize VNF reuse among the paths from the source node to each destination node. Additionally, MTSS determines the path from a source node to the central node where all VNFs can be deployed. Then it constructs the multicast tree by considering the shortest path from the last VNF location to each destination node.



(a) Net Revenue (b) Acceptance rate
Fig. 2: Performance with the increase of the destination nodes under medium-sized network



(a) Net Revenue (b) Acceptance rate
Fig. 3: Performance with the increase of the destination nodes under large-sized network

C. Results and Analysis

To verify the performance of our proposed algorithm, we compare the network net revenue and acceptance rate of six methods in a medium-sized network ($|N| = 50, |E| = 400$), respectively. In Fig.2 (a), it can be observed that all methods result in a rise in network net revenue with the increasing of destinations. The upward trend lies in that sufficient network resources guarantee more opportunities for more destinations access, gaining more network revenue. Compared to the benchmark algorithms, RSA demonstrates superior performance. On average, the network net revenue of the optimal solution is 1.07 times that of RSA.

In Fig.2 (b), with the increment of destinations, the acceptance rate of RSA and its variant algorithms goes down. The reason for the decline is that the imbalanced VNF placement strategies leads to partial network congestion, which enhancing the probability of further placement failure. Obviously, RSA outperforms other methods apart from ILP. The acceptance rate of the optimal solution is 1.02 times that of RSA. MVPboK and MTSS underperform, and indicate fluctuations in the results. This is because when their available paths from source to central node fail to meet the latency requirements or accommodate all VNFs, they prove ineffective, leading to a significantly low acceptance rate.

Furthermore, we compared the running time of all methods measured in seconds, as shown in Table II. It can be observed that RSA runs in a short time compared with MVPboK, MTSS and ILP, while it is slower than variant algorithms as expected. Especially, RSA is significantly faster than ILP, and this performance gap widens further as the problem scale

TABLE II
RUNNING TIME OF ALL METHODS IN MEDIUM-SIZE NETWORK

D/N_S	0.05	0.1	0.15	0.2	0.25
RSA	0.63	0.65	1.75	1.90	1.81
GD	0.10	0.11	0.23	0.26	0.28
RD	0.02	0.04	0.06	0.06	0.06
MVPboK	7.79	4.84	8.80	7.77	7.28
MTSS	7.36	4.75	7.41	6.07	5.87
ILP	82.17	83.60	212.43	214.50	309.03

increases. It highlights that our proposed heuristic algorithm is a promising method when the network has to rapidly obtain an effective and efficient MSFC embedding strategy according to the change of destination number.

In order to further validate the performance of our algorithm in large-sized network ($|N| = 100$, $|E| = 800$), the simulation results are shown in Fig. 3 (a) and Fig. 3 (b). It becomes evident that RSA still outperforms the benchmark algorithms, with its advantage becoming more pronounced. However, as the network scale expands, search space also increases, which in turn raises the risk of it getting trapped in local optimal. Consequently, the optimal solution achieves an average network net revenue and acceptance rate of 1.24 and 1.09 times that of RSA, respectively.

We compare the average cost for each accepted destination in SAGIN with that in the independent terrestrial network using RSA, as shown in Fig. 4. It is evident that the average cost for each accepted destination in SAGIN is reduced by 5.5%, and the blocking rate is reduced by 86.7%. This is because the wide coverage capabilities of SAGIN reduces communication hops, benefiting services with high latency requests, realizing a substantial reduction in service blocking rate.

VI. CONCLUSION

Coordination of heterogeneous objects is a key factor in resource allocation within SAGIN. Efficient MSFCs embedding strategies are essential to ensure that the SAGIN can deliver rapid and reliable global services, even when operating with limited resources. In this work, we investigate the joint optimization of VNF deployment and multicast routing in SAGIN. In order to reflect dynamic changes of physical resources in SAGIN, we set network resources and consumption coefficients for the three network segments, respectively. We establish an MSFC embedding model, considering network bandwidth capacity, node computing capacity, SFC constraints, and access control, which is formulated as an ILP problem. Considering it is NP-hard, we design a polynomial-time heuristic algorithm to solve it. The simulation results show that in medium-sized networks, our approach achieves a net revenue of 94% and an acceptance rate of 98% compared to the optimal solution. In large-sized networks, these values are 81% and 92% respectively. Although we cannot prove the optimality of our method, experiments show that it can achieve suboptimal results. Furthermore, we validate the SAGIN can reduce the deployment cost of multicast services and blocking rate, achieving an average reduction of 5.5% and 86.7% compared with independent terrestrial network, respectively.

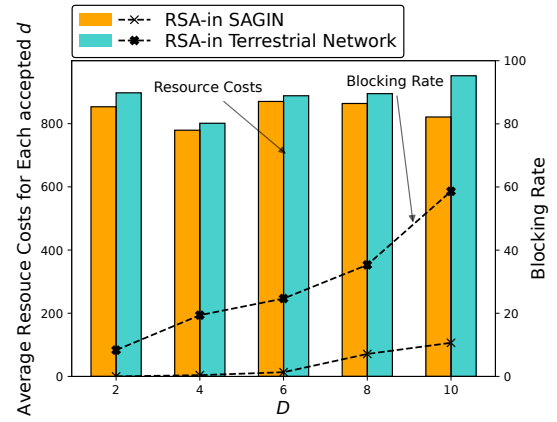


Fig. 4: Comparison between the network with integration and without integration under large-sized network.

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